



8607 - Studying cosmic noon at 200 parsec scales: resolved spectroscopy of a magnified dusty quiescent galaxy

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) 33295

ABSTRACT

We propose medium resolution NIRSpec IFU spectroscopy of A2744-33295, a gravitationally lensed galaxy at $z=2.4$. NIRCам photometry suggests A2744-33295 is a member of the growing population of reddened quiescent galaxies at cosmic noon. However, the nature of these galaxies, i.e., whether they are truly quenched, is shrouded by the dust-age degeneracy. With $\mu=18$, this source is the only known dusty galaxy at cosmic noon where spatially resolved spectroscopy is feasible on scales of hundreds of parsecs. Using spatially resolved star formation, reddening, and kinematics measurements, we will determine whether the claimed dust reddening is due to embedded star formation or is instead a sign that dust lingers in the wake of quenching and is then removed by some other channel (e.g., outflows). With A2744-33295, we will study cosmic noon at five times higher resolution than is typically achieved for local galaxies (e.g., 1 kpc for MaNGA).

OBSERVING DESCRIPTION

This program has 1 target: A2744-33295, a claimed dusty quiescent galaxy at $z=2.4$.

We will target A2744-33295 for medium resolution NIRSpec IFU spectroscopy: 11 hours with G235M/F170LP and 3 hours with G395M/F290LP. Our science goals require spatially resolved spectra with continuum $S/N > 5$. We determined the necessary exposure times by dividing the galaxy into Voronoi bins based on the NIRCам F200W imaging and modeling each bin's SED with Prospector. For each bin, we used the ETC to calculate the exposure times necessary to reach our target S/N .

No background subtraction is needed; we will estimate the background from the blank sky within the field of view. Given the source's size, no acquisition is needed. To account for leakage from open microshutter arrays, we will take short LEAKCAL exposures after each science observation; from these observations, we will mask contaminated spaxels.

We will use an 11-point dither to densely sample the IFU PSF and mitigate the effects of bad pixels.

Proposal 8607 - Targets - Studying cosmic noon at 200 parsec scales: resolved spectroscopy of a magnified dusty quiescent galaxy

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	33295	RA: 00 14 10.0890 (3.5420375d) Dec: -30 22 28.58 (-30.37461d) Equinox: J2000		
Comments: Category=Galaxy Description=[Elliptical galaxies, Starburst galaxies]					

Proposal 8607 - Observation 1 - Studying cosmic noon at 200 parsec scales: resolved spectroscopy of a magnified dusty quiescent ga...

Observation	Proposal 8607, Observation 1												Tue Mar 11 22:11:18 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
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	Comments: Category=Galaxy Description=[Elliptical galaxies, Starburst galaxies]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		MEDIUM		1		11					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G235M/F170LP	NRSIRS2	24	2	false	true	NONE	11	22	38835.625		
	2	G235M/F170LP	NRSIRS2	2	1	true	true	NONE	11	11	1765.256		
	3	G395M/F290LP	NRSIRS2	14	1	false	true	NONE	11	11	11393.923		
	4	G395M/F290LP	NRSIRS2	2	1	true	true	NONE	11	11	1765.256		